

JARPULA SAIRAM

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Education

National Institute of Technology, Patna **2023-2027**
B.Tech, Electronics and Communication Engineering *CGPA:7.01*

Sri Chaithanya Junior Collage, Yellandu **2019-2021**
State Board of Intermediate Education ,Telangana *CGPA:8.8*

Technical Skills

Languages: Verilog, System Verilog, C , basics of Python,.

Tools and Methodologies: XILINX Vivado, Cadence, MATLAB, STA, Physical Design.

Protocols: UART, SPI, I2C.

Coursework:Digital electronics , Analog electronics , Digital IC Design (prof. Janaki raman), Verilog HDL (prof. Indranil sengupta).

Experience

National Institute of Technology, Warangal **11th May 2025-10th July 2025**
Summer Internship *Warangal*

- Successfully implemented an Architectural 16-bit Approximate Radix-8 Booth Multiplier using Xilinx Vivado.
- Conducted comprehensive performance analysis focusing on power consumption, Look-Up Tables (LUTs), area, and delay, while applying the design to Sobel edge detection for image processing.

KT Semicon **21st August 2025-21st October 2025**
Internship *Hyderabad*

- Underwent intensive training in RTL design and verification methodologies using Verilog and System Verilog.
- Gained practical, hands-on experience with industry-standard tools including Xilinx Vivado and the Artix-7 FPGA board to design and test digital circuits.

National Institute of Technology Patna **10th December 2025- ongoing**
research internship. *Patna*

- Developed new phase frequency detector (PFD) circuit with 22 number of transistors. eliminate the reset path for this circuit. done PVT variation analysis.
- Do till circuit design to layout design. this circuit archive high operating frequency and phase noise with less delay.

Projects

Design and FPGA Implementation of Five Stage Pipelined RISC -V Processor. **November 2025**

- Tools: Xilinx Vivado.
- Based on the RV32I ISA using Verilog, supporting fundamental integer arithmetic, logical operations, and memory access.
- Developed advanced Hazard Detection and Forwarding units to resolve data and control dependencies, minimizing pipeline stalls and maintaining a near-ideal Cycles Per Instruction (CPI) of 1.0 for linear code execution.

Traffic light controller using FSM approach. **August 2025**

- Tools: Xilinx Vivado, Artix-7 FPGA board.
- Traffic light controller utilizing a Finite State Machine approach to manage complex signal sequences at intersections.
- Defined precise states for Red, Yellow, and Green cycles to ensure orderly traffic flow and safety at multiple directions.

Design and implement a Phase-Locked Loop (PLL) and Phase Frequency Detector (PFD). **December 2025**

- Tools: Cadence Virtuoso, 180nm.
- Design a PLL to generate a 10 MHz output synchronized with a 1 MHz reference, using a phase-frequency detector, VCO, and divide-by-10 feedback.
- Developed a new Phase Frequency Detector circuit with less number of transistors and less phase noise completed all possible analysis for this circuit.

Certifications

- NIT Warangal Summer Internship.
- KT Semicon Internship on RTL design and verification.
- Foundation Program on Nano Science and Technology from IISC Bengaluru.
- Workshop on Mastering UART Protocol from KT Semicon.
- Verilog HDL from Any skill.
- System Verilog from Any skill.
- Workshop on semiconductor technology from IIT BHU.
- Shortlisted for Samsung Fellowship cohort 7.